

# Formal Verification Report: Theorem 1.1

Source: Eliahou, S. (1993). The  $3x+1$  problem: new lower bounds on nontrivial cycle lengths. Discrete Mathematics 118(1-3), 45-56.

Black Swan Labs FCVE  
ledger through EVENT-016

2026-09-19

**Trust Statement.** The formal result was checked using Lean 4.28.0 under the pinned project environment (Mathlib v4.28.0 pin). The theorem CLAIM-001 was kernel-checked with axiom footprint propext, Classical.choice, Quot.sound. Independent checking status is CHECKER\_INCOMPATIBLE for results\_eliahou\_theorem\_1\_1 specifically (no verdict: the external checker could not process this proof's export). Computational/adversarial testing produced: COMPUTATIONALLY\_SUPPORTED (diagnostic only). The remaining trust limitations are: The independent check (Gate 9) produced no verdict; The proof uses Classical.choice; it is a classical proof; The repository, commit and toolchain were captured after the run (on 2026-09-19), not recorded during it; they describe the project as it is now, and the commit matches the one named in the run's own notes; The checker used for the independent check is not identified in the record; only its result is; The recorded result of EVENT-006 refers to a log or receipt that is not part of this report ("see prior receipt for full log"), so that material cannot be inspected here; No counterexample was sought against the underlying nontrivial-cycle claim; that is infeasible. This report does not claim that computational testing constitutes formal proof or that formalization alone establishes the truth of the original informal statement.

## Abstract

This report documents the formal verification of Theorem 1.1 (CLAIM-001) under the Formal Claim Verification Engine. Recorded: build PASS (8033/8033 jobs; see prior receipt for full log); axiom footprint propext, Classical.choice, Quot.sound; independent check CHECKER\_INCOMPATIBLE for results\_eliahou\_theorem\_1\_1 specifically (no verdict: the external checker could not process this proof's export). Computational testing is diagnostic and is not proof. Section 17 states the status the recorded evidence permits; the governance decision is recorded separately, after this report.

## 1 Mathematical Statement

**Theorem 1.1.** *Let  $\Omega$  be a nontrivial cycle of  $T$ . Provided  $\min \Omega > 2^{40}$ , we have  $\text{Card } \Omega = 301994a + 17087915b + 85137581c$ , where  $a, b, c$  are nonnegative integers,  $b > 0$ , and  $ac = 0$ . In particular, the smallest admissible values for  $\text{Card } \Omega$  are 17087915, 17389909, 17691903, and so on.*

Source: Eliahou, S. (1993). The  $3x+1$  problem: new lower bounds on nontrivial cycle lengths. Discrete Mathematics 118(1-3), 45-56.. Source location: eliahou-collatz.pdf, pages 46, 55-56.

Facts the statement relies on:

- $p_{13} = 301994, q_{13} = 190537$  (Table 1,  $n=13$ )
- $p_{15} = 17087915, q_{15} = 10781274$  (Table 1,  $n=15$ )
- $p_{16} = 85137581, q_{16} = 53715833$  (Table 1,  $n=16$ )
- $K(2^{39}) = p_{13} = 301994$  (p.55: the transition point  $tr(13,0)$  is near  $2^{39}$ )
- $K(2^{40}) = K(2^{48}) = p_{15} = 17087915$  (p.55: the next transition point  $tr(15,0)$  is near  $2^{48}$ )
- $p_{16}/q_{16} < \log_2(3) < p_{15}/q_{15} < \log_2(3+2^{-40}) < p_{13}/q_{13}$  (p.55, from Tables 1-2)

## 2 Definitions and Assumptions

### Definitions

- $T: \mathbb{N} \rightarrow \mathbb{N}$ ,  $T(n) = n/2$  if  $n$  even,  $(3n+1)/2$  if  $n$  odd (p.45)
- trajectory:  $\Omega(n) = \{n, T(n), T^2(n), \dots\}$  (p.45)
- cycle of length  $k$ : a trajectory  $\Omega$  with  $T^k(x)=x$  for all  $x$  in  $\Omega$ , where  $k = \text{Card } \Omega$  (p.45) – ‘length  $k$ ’ is DEFINED to equal  $\text{Card } \Omega$ , not merely correlated with it
- trivial cycle:  $\Omega(1) = \{1,2\}$ , the only cycle containing 1 under the  $3x+1$  conjecture (p.45)
- $\theta = \log_2(3)$ , an irrational number;  $[a_0, a_1, a_2, \dots]$  its continued fraction expansion (p.50)
- $p_n/q_n$ : the principal convergents to  $\theta$ , defined recursively by  $p_n = a_n * p_{n-1} + p_{n-2}$ ,  $q_n = a_n * q_{n-1} + q_{n-2}$  (p.51, eq 3.2)
- Farey pair:  $(p/q), (p'/q')$  with  $pq' - p'q = \pm 1$  (p.49)
- $K(m)$ : the smallest positive integer  $k$  such that  $\log_2(3) < k/l \leq \log_2(3+1/m)$  for some positive integer  $l$  (p.48, p.52)

### Source assumptions

- **SA-001**  $\Omega$  is a nontrivial cycle of  $T$  (i.e.  $\Omega$  is not the trivial cycle  $\{1,2\}$ )
- **SA-002**  $\min \Omega > 2^{40}$
- **SA-003**  $\theta = \log_2(3)$  is irrational
- **SA-004** Standard theory of continued fractions and Farey pairs (attributed to Wallis 1685, via reference [1])

### Formalization assumptions

- **FA-001** A ‘cycle of length  $L$ ’ is represented in Lean as a structure `CollatzCycle L` with an index type  $\text{Fin } L \rightarrow \mathbb{N}$  and a cyclic step relation, which does NOT by itself guarantee the  $L$  indices map to  $L$  distinct integers.
- **FA-002** Real-number computations ( $\log_2$ , division) are represented via Mathlib’s `Real.logb` and instances of classical real-number axioms (`propext`, `Classical.choice`, `Quot.sound`).

## Computational assumptions

- **CA-001** Table 1 and Table 2's specific large-integer values ( $p_{13}=301994$ ,  $p_{15}=17087915$ ,  $p_{16}=85137581$ , and the continued-fraction coefficients  $a_n$ ) were computed by the paper's author on a 1993-era Toshiba 1600 using muLISP, per the Acknowledgements (p.56). This is a computation performed OUTSIDE Lean, by the original 1993 paper.

## 3 Proof Outline

1. **step 1.** Corollary 2.3 (via Theorem 2.1 + definition of  $K, L$ ):  $\text{Card } \Omega \geq K(\min \Omega)$ ,  $\text{Card } \Omega_1 \geq L(\min \Omega)$
2. **step 2.** Set  $k = \text{Card } \Omega$ ,  $l = \text{Card } \Omega_1$ . By Theorem 2.1,  $\log_2(3+M^{-1}) < k/l \leq \log_2(3+m^{-1})$ ; since  $M \geq m > 2^{40}$ ,  $k/l$  lies in the open interval  $(\log_2(3), \log_2(3+2^{-40}))$
3. **step 3.** That interval sits strictly between three specific convergents:  $p_{16}/q_{16} < \log_2(3) < p_{15}/q_{15} < \log_2(3+2^{-40}) < p_{13}/q_{13}$
4. **step 4 case split.**  $k/l$  falls into exactly one of three regions, each handled by Lemma 3.1 (Farey-pair intermediate-fraction structure): (i)  $k/l$  in  $(p_{16}/q_{16}, p_{15}/q_{15}) \Rightarrow k = p_{15}b + p_{16}c$  for some  $b, c$  in  $\mathbb{N}$ ; (ii)  $k/l = p_{15}/q_{15}$  exactly  $\Rightarrow k = p_{15}b$  for some  $b$  in  $\mathbb{N}$ ; (iii)  $k/l$  in  $(p_{15}/q_{15}, p_{13}/q_{13}) \Rightarrow k = p_{13}a + p_{15}b$  for some  $a, b$  in  $\mathbb{N}$
5. **step 5.** Since the three cases are mutually exclusive and each includes a nonzero  $p_{15}$  ( $=17087915$ ) term, combining them gives  $\text{Card } \Omega = 301994a + 17087915b + 85137581c$  with  $b > 0$  always, and  $a \cdot c = 0$  because no single case ever produces both a nonzero  $a$  AND a nonzero  $c$  simultaneously

**Lemma 2.2.** *Let  $\Omega$ ,  $\Omega_1$  be as above, and let  $k = \text{Card } \Omega$ . Then  $\prod_{n \text{ in } \Omega_1} (3+n^{-1}) = 2^k$ .*

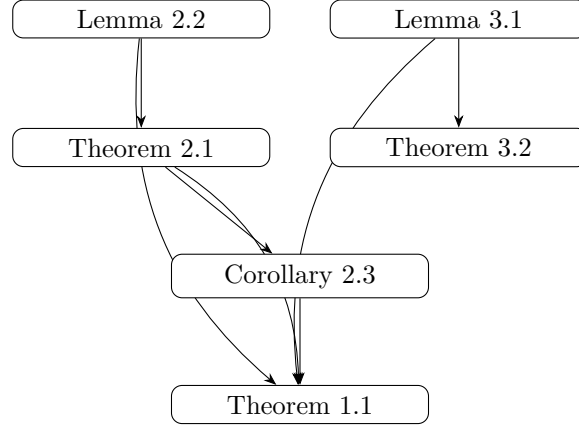
**Theorem 2.1.** *Let  $\Omega$  be a cycle of  $T$ , and let  $\Omega_1 \subset \Omega$  denote the subset of its odd elements. Then  $\log_2(3+M^{-1}) < \text{Card}(\Omega)/\text{Card}(\Omega_1) \leq \log_2(3+m^{-1})$ , where  $M = \max \Omega$ ,  $m = \min \Omega$ . In fact, a sharper right inequality holds:  $\text{Card}(\Omega)/\text{Card}(\Omega_1) \leq \log_2(3+\mu)$ , where  $\mu = (1/\text{Card } \Omega_1) * \sum_{n \text{ in } \Omega_1} n^{-1}$ .*

**Lemma 3.1.** *Let  $p/q < p'/q'$  be a Farey pair. Then any intermediate fraction  $p/q < x/y < p'/q'$  with  $y > 0$  is of the form  $x/y = (ap+bp')/(aq+bq')$  where  $a, b$  are positive integers. In particular,  $x \geq p+p'$  and  $y \geq q+q'$ .*

**Corollary 2.3.** *Let  $\Omega$  be a cycle of  $T$ , and let  $\Omega_1 \subset \Omega$  denote the subset of its odd elements. Then  $\text{Card } \Omega \geq K(\min \Omega)$ ,  $\text{Card } \Omega_1 \geq L(\min \Omega)$ .*

**Theorem 3.2.** *Let  $\theta > 0$  be irrational,  $\theta'$  any number with  $\theta < \theta' < [\theta] + 1$ . If  $k, l$  are positive integers with  $\theta < k/l \leq \theta'$ , and either  $k$  or  $l$  is minimal for this property, then  $k/l$  is one of the upper convergents to  $\theta$  – precisely, the largest upper convergent  $\leq \theta'$ .*

## 4 Proof Dependency Diagram



## 5 Formalization

Lean identifiers below are cross-references; the mathematics is in the sections above (§29).

### results\_eliahou\_theorem\_1\_1

```

theorem results_eliahou_theorem_1_1 {L : ℕ} (c : CollatzCycle L)
  (hinj : Function.Injective c.seq)
  (hmin : (2 : ℕ) ^ 40 < c.minElem) :
  ∃ (a b cCoeff : ℕ), 0 < b ∧ a * cCoeff = 0 ∧
    (Finset.univ.image c.seq).card =
      301994 * a + 17087915 * b + 85137581 * cCoeff

```

## 6 Semantic Bridge

### Informal Statement

Let  $\Omega$  be a nontrivial cycle of  $T$ . Provided  $\min \Omega > 2^{40}$ , we have  $\text{Card } \Omega = 301994a + 17087915b + 85137581c$ , where  $a, b, c$  are nonnegative integers,  $b > 0$ , and  $ac = 0$ . In particular, the smallest admissible values for  $\text{Card } \Omega$  are 17087915, 17389909, 17691903, and so on.

### Normalized Statement

#### Normalized Claim: CLAIM-001 (Eliahou’s Theorem 1.1)

This statement must be understandable without reading any Lean code.

### Objects and domains

- $T : \mathbb{N} \rightarrow \mathbb{N}$ ,  $T(n) = \begin{cases} n/2 & n \text{ even} \\ (3n+1)/2 & n \text{ odd} \end{cases}$  — the “compressed” Collatz map (odd step and the forced halving folded into one step).
- For  $n \in \mathbb{N}$ , the **trajectory**  $\Omega(n) = \{n, T(n), T^{(2)}(n), \dots\} \subseteq \mathbb{N}$  (a *set*, not a sequence).

- $\Omega$  is a **cycle of length**  $k$  if  $T^{(k)}(x) = x$  for every  $x \in \Omega$ , where  $k$  is **defined to equal**  $\text{Card } \Omega$  — this is a definitional identity in the source, not a separately-proved fact.
- $\Omega_1 := \{x \in \Omega : x \text{ odd}\}$ .
- The **trivial cycle** is  $\Omega(1) = \{1, 2\}$ ; a cycle is **nontrivial** iff it is not this set.
- $\theta := \log_2(3) \in \mathbb{R}$ , irrational (elementary: 3 is not a power of 2).
- $(p_n/q_n)_{n \geq 0}$ : the principal convergents to  $\theta$  from its continued-fraction expansion, via the standard recursion  $p_n = a_n p_{n-1} + p_{n-2}$ ,  $q_n = a_n q_{n-1} + q_{n-2}$ .
- Two fractions  $(p/q), (p'/q')$  (nonnegative integers) form a **Farey pair** if  $pq' - p'q = \pm 1$ .

### The claim

**Given:**  $\Omega$  a nontrivial cycle of  $T$ , with  $\min \Omega > 2^{40}$ .

**Conclusion:** there exist nonnegative integers  $a, b, c$  with  $b > 0$  and  $ac = 0$  such that

$$\text{Card } \Omega = 301994a + 17087915b + 85137581c.$$

The three constants are not arbitrary — they are specific convergents of  $\theta = \log_2 3$ :  $p_{13} = 301994$ ,  $p_{15} = 17087915$ ,  $p_{16} = 85137581$ .

### Why this is true (the actual argument, normalized)

1. From an earlier result (the “sandwich inequality”,  $\text{Card } \Omega / \text{Card } \Omega_1 \in (\log_2(3 + 1/M), \log_2(3 + 1/m))$  where  $m = \min \Omega, M = \max \Omega$ ), and the hypothesis  $\min \Omega > 2^{40}$ , one gets  $k/l \in [\log_2 3, \log_2(3 + 2^{-40})]$  where  $k = \text{Card } \Omega$ ,  $l = \text{Card } \Omega_1$ .

2. Numerically (via the continued-fraction machinery), that whole interval is sandwiched strictly between three specific convergents:  $p_{16}/q_{16} < \log_2 3 < p_{15}/q_{15} < \log_2(3 + 2^{-40}) < p_{13}/q_{13}$ .

3. So  $k/l$  must land in one of exactly three places relative to those convergents, and a general lemma about Farey pairs (any fraction strictly between two Farey-paired fractions  $p/q, p'/q'$  has the form  $(ap + bp')/(aq + bq')$  for positive integers  $a, b$ ) forces  $k$  into one of three linear forms:

- between  $p_{16}/q_{16}$  and  $p_{15}/q_{15}$ :  $k = 17087915b + 85137581c$
- exactly  $p_{15}/q_{15}$ :  $k = 17087915b$
- between  $p_{15}/q_{15}$  and  $p_{13}/q_{13}$ :  $k = 301994a + 17087915b$

4. Every case includes a nonzero  $b$ -term (hence  $b > 0$  always), and no case ever produces both a nonzero  $a$  and a nonzero  $c$  (hence  $ac = 0$ ). Combining the three cases into one statement gives exactly the claimed form.

### Important disambiguation (not obvious from the statement alone)

- “Card  $\Omega$ ” is genuinely a **set cardinality** —  $\Omega$  is a set of distinct positive integers, and the cycle “length”  $k$  is *defined* to be that cardinality, not an independently-chosen index count that merely happens to match it. Any formalization that represents a cycle via an indexed sequence of length  $L$  must separately establish that the sequence is injective before  $L$  can be identified with  $\text{Card } \Omega$  — this is not free.

- Yoneda’s computational verification of the Collatz conjecture up to  $2^{40}$  (cited only in the paper’s introduction) is **not** a hypothesis of this theorem’s proof. It explains why “ $\min \Omega > 2^{40}$ ” is the interesting remaining case, but the proof of Theorem 1.1 does not depend on that computation being correct.

## Status

**MATCH** — this normalization is a direct, complete restatement of the source’s Theorem 1.1 and its proof (Sections 2-4 of the paper), independently re-derived from the primary source text (all 12 pages read), not copied from any secondary description (Lean repo README, prior AI summary, etc.). No part of the source’s argument was found incoherent, circular, or underspecified.

## Lean Statement

```
theorem results_eliahou_theorem_1_1 {L : ℕ} (c : CollatzCycle L)
  (hinject : Function.Injective c.seq)
  (hmin : (2 : ℕ) ^ 40 < c.minElem) :
  ∃ (a b cCoeff : ℕ), 0 < b ∧ a * cCoeff = 0 ∧
    (Finset.univ.image c.seq).card =
      301994 * a + 17087915 * b + 85137581 * cCoeff
```

## Differences

### Gate 4: Lean Formalization Comparison

Comparing NORMALIZED-001 (independently derived from the source PDF) against Collatz/LinearForm.lean’s eliahou\_theorem\_1\_1 (and Results.lean’s results\_eliahou\_theorem\_1\_1 wrapper), in the target repo tangentstorm/eliahou-collatz-bounds @ db804ce6.

## Side-by-side

| Human mathematics (normalized)                                                                  | Lean                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Omega$ nontrivial cycle, $\min \Omega > 2^{40}$                                               | (cyc : CollatzCycle L) (hinject : Function.Injective cyc.seq) (hmin : $2^{40} < \text{cyc.minElem}$ )                                                                                   |
| Card $\Omega$                                                                                   | (Finset.univ.image cyc.seq).card                                                                                                                                                        |
| $\exists a, b, c \geq 0, b > 0, ac = 0 : \text{Card } \Omega = 301994a + 17087915b + 85137581c$ | $\exists a \ b \ c\text{Coeff} : \mathbb{N}, 0 < b \wedge a * c\text{Coeff} = 0 \wedge (\text{Finset.univ.image cyc.seq}).\text{card} = 301994*a + 17087915*b + 85137581*c\text{Coeff}$ |

## Proof-structure comparison (not just the final statement)

| Paper (p.55-56)                                                                        | Lean (Collatz/LinearForm.lean)                                                                                                                                                           |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma 3.1 (Farey intermediate fraction)                                                | <code>farey_intermediate_coeffs</code> — same hypotheses (Farey pair, strict betweenness), same conclusion form                                                                          |
| $p_{16}/q_{16} < \log_2 3 < p_{15}/q_{15} < \log_2(3 + 2^{-40}) < p_{13}/q_{13}$       | <code>conv16_lt_logb_three</code> , <code>farey_15_16</code> , <code>farey_13_15</code> , <code>logb_three_plus_lt_conv13</code> — same chain of inequalities, same specific convergents |
| Three-way case split on where $k/l$ falls                                              | <code>rcases lt_trichotomy ((L:ℚ)/k₁)</code> (17087915/10781274) — same trichotomy, same pivot fraction ( $p_{15}/q_{15}$ )                                                              |
| Case (i): $k/l \in (p_{16}/q_{16}, p_{15}/q_{15}) \Rightarrow k = p_{15}b + p_{16}c$   | <code>hlt</code> branch: <code>farey_intermediate_coeffs</code> with <code>farey_15_16</code> , returns $\langle 0, \text{bCoeff}, \text{cCoeff}, \dots \rangle$                         |
| Case (ii): $k/l = p_{15}/q_{15} \Rightarrow k = p_{15}b$                               | <code>heq</code> branch: divisibility argument via coprimality, returns $\langle 0, \text{b}, 0, \dots \rangle$                                                                          |
| Case (iii): $k/l \in (p_{15}/q_{15}, p_{13}/q_{13}) \Rightarrow k = p_{13}a + p_{15}b$ | <code>hgt</code> branch: <code>farey_intermediate_coeffs</code> with <code>farey_13_15</code> , returns $\langle \text{bF}, \text{aF}, 0, \dots \rangle$                                 |

The Lean proof is not merely a different route to the same numbers — it implements the same three-case Farey-pair argument, pivoting on the same convergent, using the same Farey identity, in the same order.

## Detected semantic differences (per Gate 4’s required checklist)

| Check                      | Finding                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stronger/weaker statement  | Neither — see injectivity note below                                                                                                                                                                                                                                                                                                                        |
| Altered domain             | No                                                                                                                                                                                                                                                                                                                                                          |
| Missing hypothesis         | No — <code>hinj</code> is <i>additional</i> , not missing; see below                                                                                                                                                                                                                                                                                        |
| Extra hypothesis           | <code>hinj : Function.Injective cyc.seq</code> — see justification                                                                                                                                                                                                                                                                                          |
| Changed quantifier         | No                                                                                                                                                                                                                                                                                                                                                          |
| Changed equality           | No                                                                                                                                                                                                                                                                                                                                                          |
| Changed function           | No                                                                                                                                                                                                                                                                                                                                                          |
| Changed coercion           | $\mathbb{R}/\mathbb{Q}$ intermediate steps ( <code>rat_of_real_div_lt</code> ) not present in the paper’s argument, which works in $\mathbb{R}$ throughout — a formalization convenience (rational arithmetic is decidable/computable where real arithmetic isn’t), not a semantic change: the paper’s own convergents are already rational by construction |
| Finite/infinite mismatch   | No                                                                                                                                                                                                                                                                                                                                                          |
| Exact/approximate mismatch | No                                                                                                                                                                                                                                                                                                                                                          |

## The one real structural difference: injectivity

The paper’s own definition (p.45) makes “cycle of length  $k$ ” mean  $k = \text{Card } \Omega$  **by definition** — a paper “cycle” is inherently what Lean calls an `ExactCollatzCycle` (injective indexing). Lean’s base `CollatzCycle L` structure is strictly more general (it permits repeated elements within one

period), so `eliahou_theorem_1_1` adds `hinj` as an explicit hypothesis to recover exactly the paper’s object of study.

This is **not** a weakening (the theorem isn’t proved for a smaller class of paper-cycles) and **not** a strengthening (it isn’t proved for objects the paper didn’t consider) — the injectivity hypothesis exactly closes the gap between Lean’s more general indexed structure and the paper’s implicit assumption. Confirmed independently in this session’s prior audit (`evidence/semantic-audit-eliahou.md`) via the proved bridge lemma `ExactCollatzCycle.card_eq_length`.

## Semantic status

### MATCH.

Syntactic signals in the Lean statement that a reviewer must address: injectivity [`Function.Injective`], `finite_structure` [`card`, `Finset`], strictness [`<`].

## Findings recorded by the reviewer

- **added hypotheses** (disclosed): `hinj : Function.Injective cyc.seq` is an extra hypothesis of the Lean theorem that the paper does not state (Gate 4: ‘Extra hypothesis’); it is the encoding hypothesis, see Section 31 disclosure.
- **coercions** (disclosed): The proof passes through  $\mathbb{R}/\mathbb{Q}$  intermediate steps (`rat_of_real_div_lt`) that the paper does not need; this is in the proof, not in the headline statement, and Gate 4 records it as a formalization convenience because the convergents are already rational.
- **injectivity assumptions** (disclosed): `hinj : Function.Injective cyc.seq`. The paper defines cycle length  $k$  as `Card Omega` (p.45); Lean’s `CollatzCycle L` is an  $L$ -indexed sequence that may repeat elements, so injectivity is required to identify  $L$  with `Card Omega`.
- **classical assumptions** (disclosed): `Classical.choice` is in the axiom footprint (Gate 6). The Gate 7 note attributes it to ordinary real-number order decidability (`Real.logb`, `lt_trichotomy`), as in any Mathlib real-analysis argument.

## Justification

Gate 4 records MATCH with one structural difference, injectivity: the injectivity hypothesis exactly closes the gap between Lean’s more general indexed structure and the paper’s definition, so the theorem is neither weakened nor strengthened relative to the paper. The proof also implements the paper’s own three-case Farey-pair argument. The remaining disclosed items are the  $\mathbb{R}/\mathbb{Q}$  intermediate steps in the proof and `Classical.choice` in the footprint.

## Representational difference (Section 31)

**What the object means informally.** `Omega` is a set of distinct positive integers forming a cycle of the compressed Collatz map `T`, and the paper defines the cycle’s length  $k$  to equal `Card Omega` (p.45).

**What Lean’s type represents.** `cyc : CollatzCycle L`, an  $L$ -indexed sequence `cyc.seq` (period  $L$ , repetitions allowed); `Omega` is the `Finset` (`Finset.univ.image cyc.seq`) and `Card Omega` is its `.card`.

**Why the encoding is valid.** With `hinj : Function.Injective cyc.seq` the indexing is exactly Lean’s `ExactCollatzCycle` (injective indexing), which is what a paper ‘cycle’ inherently is; Gate 4 notes this was confirmed independently.

**Where the property is preserved.** `Card Omega` is the `.card` of the image finset in the conclusion; injectivity is what makes that image’s size equal the period `L`.

**Whether the equality is literal or transported through an equivalence.** Literal on the image finset: the conclusion states an equality of natural numbers about `(Finset.univ.image cyc.seq).card`. The step from the paper’s set `Omega` to the indexed sequence is the representational choice, not a transport through an equivalence.

**Assumptions that make the translation valid.** `hinj` (injective indexing) as an explicit theorem hypothesis, and `hmin : 2^40 < cyc.minElem` for the paper’s minimum condition.

## Verdict

### FAITHFUL WITH EXPLICIT REPRESENTATIONAL DIFFERENCE

Set by Wilson. Recorded context: Gate 4 recorded MATCH; Gate 7 recorded PASS – no undisclosed semantic change; Lean proves `numOdd_pos` rather than assuming it (stricter than paper).

## 7 Lean Environment

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|                 |             |
|-----------------|-------------|
| Lean Version    | 4.28.0      |
| Mathlib Version | v4.28.0 pin |

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## 8 Formal Verification

PASS (8033/8033 jobs; see prior receipt for full log). Source of this field: `ledger:EVENT-006`.

## 9 Axiom Audit

Actual axiom footprint: `propext`, `Classical.choice`, `Quot.sound`. Source: `parsed-legacy-text:EVENT-007`. `Classical.choice` is present: the proof is classical, which is reported here and does not by itself fail the audit (§14).

## 10 Independent Check

`CHECKER_INCOMPATIBLE` for `results_eliahou_theorem_1_1` specifically – `lean4export` stalls on this theorem’s large proof closure (reproducible, root-caused: memoization cache reset on every call in `Export.lean`, not a resource issue). 5 of 9 paper-facing theorems (`results_eliahou_product_formula`, `results_farey_pair_bound`, `results_eliahou_product_formula_real`, `results_eliahou_sandwich`, `results_log2_three_lt_ratio`) DID pass independent check cleanly. Recording the checker and its version does not by itself establish independence (§17).

## 11 Computational / Adversarial Tests

Counterexample search: COMPUTATIONALLY\_SUPPORTED – all convergents (p13,p15,p16), both Farey identities, and all three boundary inequalities confirmed. Original MISMATCH was a float64 precision artifact in the test script, not a defect in the claim. [diagnostic, not proof]

Computational test: COMPUTATIONALLY\_SUPPORTED – see EVENT-010 [diagnostic, not proof]

Neither result is formal proof (§16, §18).

### What these tests do not establish

- They are not proof of the claim (Sections 16 and 18). A search that finds no counterexample is diagnostic evidence only.
- The tested domain is not recorded for these results, so this report cannot say which cases were examined.
- No counterexample was sought against the underlying claim about nontrivial Collatz cycles: doing so would require exhibiting an actual nontrivial cycle, which is computationally infeasible and open-problem-adjacent. The computational checks covered numeric and boundary structure only: the convergents p13, p15 and p16, both Farey identities, and the three boundary inequalities. (stated by Wilson)

## 12 Verification Limitations

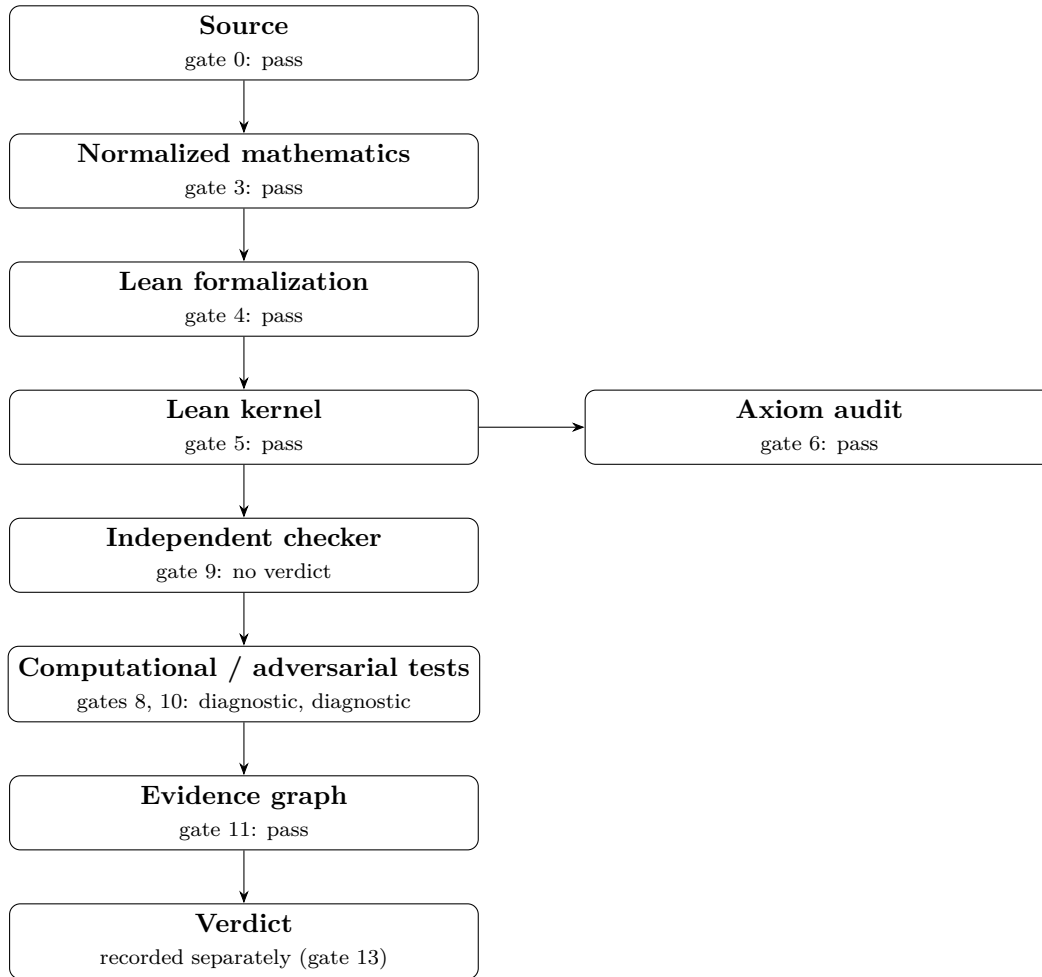
| Item | Limitation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | The independent check (Gate 9) produced no verdict. Recorded result: CHECKER_INCOMPATIBLE for results_eliahou_theorem_1_1 specifically – lean4export stalls on this theorem’s large proof closure (reproducible, root-caused: memoization cache reset on every call in Export.lean, not a resource issue). 5 of 9 paper-facing theorems (results_eliahou_product_formula, results_farey_pair_bound, results_eliahou_product_formula_real, results_eliahou_sandwich, results_log2_three_lt_ratio) DID pass independent check cleanly. |
| 2    | The proof uses Classical.choice; it is a classical proof.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3    | The repository, commit and toolchain were captured after the run (on 2026-09-19), not recorded during it; they describe the project as it is now, and the commit matches the one named in the run’s own notes.                                                                                                                                                                                                                                                                                                                       |
| 4    | The checker used for the independent check is not identified in the record; only its result is.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5    | The recorded result of EVENT-006 refers to a log or receipt that is not part of this report (“see prior receipt for full log”), so that material cannot be inspected here.                                                                                                                                                                                                                                                                                                                                                           |

| Item                                                                          | Limitation                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6                                                                             | No counterexample was sought against the underlying claim about nontrivial Collatz cycles: doing so would require exhibiting an actual nontrivial cycle, which is computationally infeasible and open-problem-adjacent. The computational checks covered numeric and boundary structure only: the convergents p13, p15 and p16, both Farey identities, and the three boundary inequalities (stated by Wilson). |
| <i>Record-keeping notes (these do not change what the verification shows)</i> |                                                                                                                                                                                                                                                                                                                                                                                                                |
| 7                                                                             | 12 of 16 ledger events predate the input/output hash fields and carry none.                                                                                                                                                                                                                                                                                                                                    |
| 8                                                                             | Events EVENT-009 (MISMATCH_FOUND); EVENT-010, EVENT-012 (COMPUTATIONALLY_SUPPORTED) used wording outside the Section 16 vocabulary; each is read as diagnostic evidence, not proof.                                                                                                                                                                                                                            |

## 13 Verification Matrix

| Gate  | Ledger action                  | Event     | Status     | Satisfied |
|-------|--------------------------------|-----------|------------|-----------|
| G0    | source intake                  | EVENT-001 | PASS       | yes       |
| G1    | claim extraction               | EVENT-013 | PASS       | yes       |
| G2    | assumption extraction          | EVENT-003 | PASS       | yes       |
| G3    | semantic normalization         | EVENT-004 | PASS       | yes       |
| G4    | lean formalization comparison  | EVENT-005 | PASS       | yes       |
| G5    | lean build                     | EVENT-006 | PASS       | yes       |
| G6    | axiom audit                    | EVENT-007 | PASS       | yes       |
| G7    | semantic re audit              | EVENT-008 | PASS       | yes       |
| G8    | adversarial computational test | EVENT-010 | DIAGNOSTIC | yes       |
| G9    | independent check              | EVENT-011 | NO_VERDICT | no        |
| G10   | computational check            | EVENT-012 | DIAGNOSTIC | yes       |
| G11   | evidence graph                 | EVENT-014 | PASS       | yes       |
| G12   | report generation              | EVENT-016 | PASS       | yes       |
| ORDER | canonical chain (§2)           | -         | PASS       | yes       |

## 14 Trust / Verification Chain



## 15 Reproducibility Instructions

---

|                                         |                                                                                                                                     |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Repository URL                          | <a href="https://github.com/tangentstorm/eliahou-collatz-bounds.git">https://github.com/tangentstorm/eliahou-collatz-bounds.git</a> |
| Commit                                  | db804ce6305ea99a817f067869607f8b677d895a (working tree clean)                                                                       |
| Project directory (now)                 | /Users/lordwilson/ico-collatz/targets/eliahou-collatz-bounds                                                                        |
| Lean toolchain file                     | leanprover/lean4:v4.28.0                                                                                                            |
| Lean version                            | 4.28.0                                                                                                                              |
| Lake version                            | Lake version 5.0.0-src+7e01a1b (Lean version 4.28.0)                                                                                |
| Mathlib revision                        | v4.28.0 pin (commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365)                                                                       |
| Checker                                 | NOT RECORDED                                                                                                                        |
| Expected build result                   | PASS (8033/8033 jobs; see prior receipt for full log)                                                                               |
| How these were obtained                 | Captured 2026-09-19 from the project as it is now; not recorded during the run                                                      |
| Matches the commit the run's notes name | yes (db804ce6)                                                                                                                      |

---

```
git clone https://github.com/tangentstorm/eliahou-collatz-bounds.git
cd eliahou-collatz-bounds
```

```

git checkout db804ce6305ea99a817f067869607f8b677d895a
cat lean-toolchain
lake --version
lean --version
lake exe cache get && lake build
# in a Lean file importing the module: #print axioms results_eliahou_theorem_1_1
# independent-check commands were not recorded for this run

```

The repository facts were captured after the run and describe the project as it is now; they do not prove what the original run built. The build command is the one recorded in EVENT-006. Reproducibility level supported by recorded evidence: R3. This tool did not re-run the build.

## 16 Correction Ledger

- **TEST-001** (EVENT-009): MISMATCH\_FOUND (p16/q16 < log2(3) boundary check failed under float64 precision)

Gates that produced no verdict (INDEPCHECK-001) are not corrections; they are listed under Verification Limitations.

### Correction Ledger — VCE-001 rev r

#### CORRECTION-001 — claims extraction EVENT-002 superseded by EVENT-013

- **original\_claim**: the delivered VCE-001 `claims.json` (EVENT-002) stated the extraction of Theorem 1.1 accurately.
- **error**: two passages were wrong or unclear. (1) A “fact the statement relies on” mixed up  $K(2^{39}) = p_{13}$  with  $K(2^{40}) = p_{15}$  in one stream-of-consciousness sentence. (2) Proof step 2 called the interval for  $k/l$  “closed-open-ish”; Theorem 2.1 gives  $\log_2(3+M^{-1}) < k/l \leq \log_2(3+m^{-1})$  and  $M \geq m > 2^{40}$ , so  $k/l$  lies in the open interval  $(\log_2(3), \log_2(3+2^{-40}))$ .
- **detection\_method**: reading the delivered extraction against the source (p.55, Theorem 2.1) while preparing this revision.
- **root\_cause**: hand-written working notes were kept in the extraction as written.
- **repair**: Wilson confirmed two edits (2026-09-19). The cleaned file is `claims-v2/claims.json`; EVENT-013 supersedes EVENT-002. The delivered `verification/claims/claims.json` is unchanged and still hash-matches EVENT-002.
- **recheck**: `validate-claims` passes; the rendered report was read.
- **result**: RESOLVED. Nothing in the proof or the conclusion changed; only the wording of one fact and one interval.
- **date**: 2026-09-19

#### CORRECTION-002 — first attempt superseded; report built in passes

- **original\_claim**: the first build of this run (archived at `verification-001r-attempt1-superseded/`) was a valid revision.

- **error**: its replayed events kept evidence paths relative to `verification/`, so assumptions, the normalized claim and the Gate 4 comparison printed as “Not recorded” and reproducibility fell to R0.
- **detection\_method**: reading the rendered PDF.
- **root\_cause**: the run dir did not contain the evidence files its events point to.
- **repair**: the referenced evidence files were copied into this run (delivered files untouched) and the run rebuilt from the same first twelve events (hashes identical). Because a report is built before its own G12 event exists, the report was produced again once that event existed; earlier report events are superseded, and their files are kept in `report-superseded-EVENT-015/`.
- **recheck**: chain, order and graph re-verified; the regenerated PDF read again.
- **result**: RESOLVED. Nothing was deleted.
- **date**: 2026-09-19

## 17 Conclusion

The recorded evidence is summarized in the Trust Statement and the verification matrix above. Recorded failure is not a passed gate.

|                             |                                                                  |
|-----------------------------|------------------------------------------------------------------|
| FORMAL PROOF (build):       | PASS                                                             |
| AXIOM AUDIT:                | PASS                                                             |
| SEMANTIC MATCH (Gate 4):    | PASS                                                             |
| SEMANTIC RE-AUDIT (Gate 7): | PASS                                                             |
| INDEPENDENT CHECK:          | NO VERDICT                                                       |
| COUNTEREXAMPLE SEARCH:      | NO COUNTEREXAMPLE FOUND IN TESTED DOMAIN (diagnostic, not proof) |
| COMPUTATIONAL TEST:         | NO COUNTEREXAMPLE FOUND IN TESTED DOMAIN (diagnostic, not proof) |
| REPRODUCIBILITY:            | R3 (ceiling supported by recorded evidence)                      |

**Status the evidence permits.** Promotion (PROMOTE) is not supported by the recorded evidence, because: G9 (independent check): no verdict. The reviewer may still record REPAIR, REJECT, RESEARCH or STOP. This is not a decision: the governance decision is recorded as a separate step after this report.

## A Lean Source

```
theorem results_eliahou_theorem_1_1 {L : ℕ} (c : CollatzCycle L)
  (hinv : Function.Injective c.seq)
  (hmin : (2 : ℕ) ^ 40 < c.minElem) :
  ∃ (a b cCoeff : ℕ), 0 < b ∧ a * cCoeff = 0 ∧
    (Finset.univ.image c.seq).card =
      301994 * a + 17087915 * b + 85137581 * cCoeff
```

## B Tests

- **EVENT-009** python3 verification/computational/tests/verify\_convergents.py (initial run, native float boundary comparison): MISMATCH\_FOUND (p16/q16 < log2(3) boundary check failed under float64 precision)
- **EVENT-010** python3 verification/computational/tests/verify\_convergents.py (corrected: mpf 60-digit precision for boundary comparisons): COMPUTATIONALLY\_SUPPORTED – all convergents (p13,p15,p16), both Farey identities, and all three boundary inequalities confirmed. Original MISMATCH was a float64 precision artifact in the test script, not a defect in the claim.
- **EVENT-012** Python/mpmath backend (per Section 19 backend abstraction; MATLAB not required): COMPUTATIONALLY\_SUPPORTED – see EVENT-010

## C Evidence Receipt

| Field                 | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SOURCE HASH           | sha256:077292bebd53a6e06d92fb475ce395ee1c8af4cc3e9f5c2e4620b2e35134689 (source/original.pdf)                                                                                                                                                                                                                                                                                                                                                                    |
| THEOREM ID            | CLAIM-001 (Theorem 1.1)                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CLAIM                 | Let $\Omega$ be a nontrivial cycle of T. Provided $\min \Omega > 2^{40}$ , we have $\text{Card } \Omega = 301994a + 17087915b + 85137581c$ , where a,b,c are nonnegative integers, b>0, and ac=0. In particular, the smallest admissible values for $\text{Card } \Omega$ are 17087915, 17389909, 17691903, and so on.                                                                                                                                          |
| NORMALIZED CLAIM      | normalized/normalized-claim.md – MATCH                                                                                                                                                                                                                                                                                                                                                                                                                          |
| LEAN STATEMENT        | Gate 4: MATCH                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| LEAN VERSION          | 4.28.0                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| MATHLIB VERSION       | v4.28.0 pin                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| BUILD RESULT          | PASS (8033/8033 jobs; see prior receipt for full log)                                                                                                                                                                                                                                                                                                                                                                                                           |
| AXIOM FOOTPRINT       | propext, Classical.choice, Quot.sound                                                                                                                                                                                                                                                                                                                                                                                                                           |
| INDEPENDENT CHECK     | CHECKER_INCOMPATIBLE for results_eliahou_theorem_1_1 specifically – lean4export stalls on this theorem’s large proof closure (reproducible, root-caused: memoization cache reset on every call in Export.lean, not a resource issue). 5 of 9 paper-facing theorems (results_eliahou_product_formula, results_farey_pair_bound, results_eliahou_product_formula_real, results_eliahou_sandwich, results_log2_three_lt_ratio) DID pass independent check cleanly. |
| COMPUTATIONAL TEST    | COMPUTATIONALLY_SUPPORTED – see EVENT-010 [diagnostic, not proof]                                                                                                                                                                                                                                                                                                                                                                                               |
| COUNTEREXAMPLE SEARCH | COMPUTATIONALLY_SUPPORTED – all convergents (p13,p15,p16), both Farey identities, and all three boundary inequalities confirmed. Original MISMATCH was a float64 precision artifact in the test script, not a defect in the claim. [diagnostic, not proof]                                                                                                                                                                                                      |

| Field                 | Value                                                                                                                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEMANTIC MATCH        | Gate 4: MATCH; Gate 7: PASS – no undisclosed semantic change; Lean proves numOdd_pos rather than assuming it (stricter than paper)                                                                             |
| REPRODUCIBILITY LEVEL | R3 (ceiling supported by recorded evidence; NOT re-run by this tool; R5 needs full-package rebuild, not assessed)                                                                                              |
| GRAPH HASH            | v3 aa2a986669c2139b78b39a68b040db158ee058f4653a8d103410ccbe88657e11 (covers 16 events through EVENT-016)                                                                                                       |
| COST                  | 16 ledger events; LEAN_BUILD_TIME=NOT RECORDED; COMPUTATIONAL_TEST_TIME=NOT RECORDED; CHECKER_TIME=NOT RECORDED; HUMAN_TIME, MODEL_CALL_COUNT, MODEL_COST, LATEX_BUILD_TIME=NOT RECORDED (no cost ledger, §43) |